

23CSE473 Neural Networks and Deep Learning Group 9

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1. Reference for

a. Dataset:

https://figshare.com/articles/dataset/TriModal_Ripeness_6/30783827/1?file=60094544

2. Problem Statement

Explainable Out-of-Distribution Detection for Foreign Object Rejection in Automated Fruit and Vegetable Grading Systems

Automated fruit and vegetable grading systems rely on vision models trained to recognize a fixed set of known produce categories. In real-world deployment, however, these systems inevitably encounter foreign objects and unrecognized items that were never part of the training distribution packaging debris, contaminants, or produce varieties the model has never seen which a standard closed-set classifier will confidently but incorrectly assign to one of its known classes rather than flagging as anomalous. This project proposes an explainable out-of-distribution (OOD) detection framework for automated fruit and vegetable grading systems, designed to reject such unrecognized inputs rather than misclassify them, while also providing visual justification for every accept/reject decision.

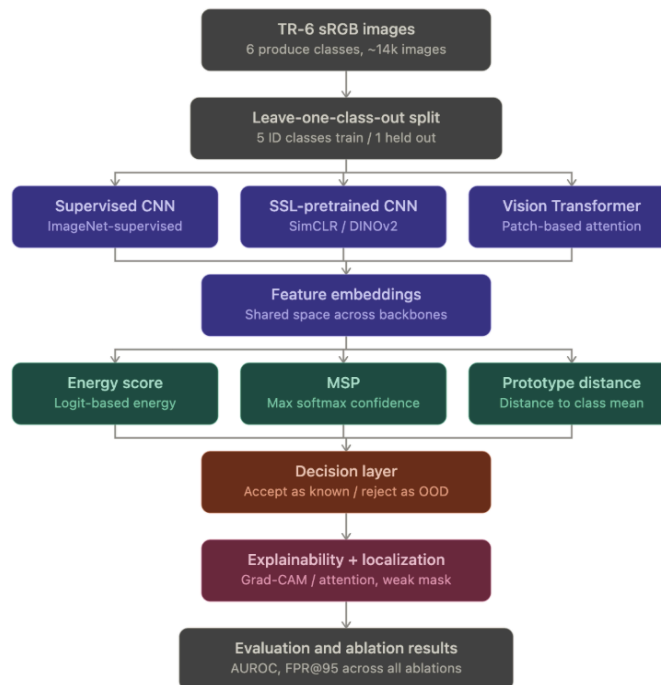
Using the TR-6 (TriModal Ripeness 6) dataset comprising sRGB images of six produce classes (guava, carrot, tomato, Indian gooseberry, banana, mango) the system is trained on five known classes while the sixth is held out entirely and used at test time as a proxy for a novel, unrecognized object, following the standard leave-one-class-out semantic OOD evaluation protocol, rotated across all six classes for robust evaluation. The framework further investigates how representation quality (supervised vs. self-supervised pretraining, CNN vs. Vision Transformer backbones) and choice of OOD scoring function (energy score, maximum softmax probability, prototype distance) affect detection reliability, and integrates Grad-CAM/attention-based explainability doubling as weak, annotation-free localization of the anomalous region to make every rejection decision interpretable rather than a black-box output.

3. Dataset Description

- a. Number of images: Approximately 29,000 images (combined sRGB and IR-Fusion thermal images)
- b. Size: 18.99 GB
- c. Type of image:
 - JPG images consisting of:
 - sRGB (RGB color) images
 - IR-Fusion thermal images (thermal images fused with visible spectrum)

4. Proposed methodology

- a. Architecture Diagram:



- b. Justification/ need for each module:

1. TR-6 sRGB subset

This is the input data layer, scoped deliberately to the sRGB modality of TR-6 rather than the full trimodal dataset. sRGB is sufficient to test the OOD detection and explainability methodology this project is built around, and keeping scope to one modality keeps model training, ablations, and Grad-CAM interpretation tractable within a semester timeline.

2. Leave-one-class-out split

This is the foundation the entire project depends on. Without holding out a class entirely from training, there is no genuine "unrecognized object" to test detection against the task would collapse into ordinary closed-set classification, and the "OOD" claim in the problem statement would have nothing behind it. Rotating the held-out class across all 6 categories (rather than fixing one) and averaging results avoids the result looking cherry-picked to a single arbitrary split.

3. Supervised CNN

Serves as the baseline backbone a standard ImageNet-pretrained CNN fine-tuned on the 5 in-distribution classes. Necessary as the reference point every subsequent comparison (SSL backbone, ViT, scoring methods) is measured against.

4. SSL-pretrained CNN (SimCLR/DINOv2)

Tests whether self-supervised representations separate in-distribution and out-of-distribution samples better than purely supervised ones. Supervised training optimizes features to collapse toward class decision boundaries, which can discard information irrelevant to classification but useful for detecting novelty; self-supervised pretraining tends to preserve richer, more general features, which is why it's included as a direct comparison rather than assumed.

5. Vision Transformer

Tests whether a fundamentally different inductive bias global self-attention instead of a CNN's local receptive field improves the model's ability to recognize "this doesn't fit the pattern I know." Also provides a second, independent explainability signal (raw attention maps) to compare against Grad-CAM later in the pipeline.

6. Feature embeddings (merge point)

This is the architectural seam that makes the backbone comparison fair and clean: regardless of which backbone produced them, embeddings land in a common space so the downstream OOD scoring and decision logic don't need to change per backbone. Without this merge point, you'd effectively be building three disconnected pipelines rather than one pipeline with a swappable, comparable component.

7. Energy score

A theoretically grounded OOD scoring method derived from the log-partition function over the model's logits. Included because it consistently outperforms simple softmax-based confidence scores in prior OOD detection literature, making it a strong primary candidate.

8. MSP (Maximum Softmax Probability)

The simplest possible confidence-based OOD score just the top softmax output.

Included specifically because it's the standard baseline every OOD paper reports against; without it, there's no reference point to show whether the energy score or prototype distance are actually improvements.

9. Prototype distance

A feature-space scoring method (distance to the nearest class-mean embedding) that doesn't depend on the classifier's softmax calibration at all. Included as a check against cases where softmax-based scores are unreliable or overconfident a known failure mode of neural network classifiers on unfamiliar inputs.

10. Decision layer

Converts whichever OOD score is used into an actual accept/reject outcome via a threshold. Kept as a distinct module because the threshold is a tunable operating point reported across its full range via AUROC rather than fixed to one arbitrary cutoff, since a real grading line would need to choose this threshold based on acceptable false-rejection cost.

11. Explainability + localization

Grad-CAM (for CNN backbones) or attention rollout (for the ViT backbone) shows *why* the decision layer flagged an input as unfamiliar, which is what makes this an "Explainable" OOD system rather than a black-box detector directly satisfying the problem statement's core requirement. Thresholding the resulting heatmap into a binary mask further provides a weakly-supervised approximation of defect/foreign-object localization, without requiring manually annotated bounding boxes that don't exist in this dataset.

12. Evaluation and ablation results

Aggregates AUROC and FPR@95 across all 6 leave-one-class-out folds and both ablation axes (backbone × scoring method). This is where the ablation design actually pays off in the report instead of a single detection number, it produces a comparative analysis of what actually drives OOD detection performance in this domain (representation quality vs. scoring rule), which is the difference between an engineering exercise and a research contribution.

5. Check the dataset

5.1 Preprocessing checks needed on the dataset

1. Class imbalance confirmed, not just a risk

The per-category counts in the paper show real imbalance: Tomato totals ~4,488 sRGB images (2,244 Normal + 2,244 Classified combined) while Banana totals only ~646 (323 Normal + 323 Classified). Gooseberry sits around 2,822. This imbalance will bias both the classifier's decision boundaries and your OOD score

thresholds toward over-represented classes unless addressed via class-weighted loss, oversampling the minority classes, or capping the majority classes during training.

2. Same-specimen leakage across splits high risk given the capture protocol

Each fruit was photographed 3 times a day (16 side-angle + 1 top-down per session) over its entire decomposition period. This means many images in the dataset are the *same physical fruit* at different angles/times, not independent samples. If your train/val/test (or fold) split is done by image rather than by specimen, the model can memorize a specific fruit's blemishes rather than learning general class features inflating both classification accuracy and OOD scores artificially. You need to split by specimen/session ID, not by individual image, using the timestamp in the filenames to group images from the same fruit together.

3. Normalization strategy differs per backbone branch

Standard ImageNet mean/std normalization for the supervised CNN and ViT branches; the SSL branch (SimCLR/DINOv2) typically needs its own augmentation pipeline (random crop, color jitter, Gaussian blur) rather than reusing the supervised branch's preprocessing worth stating explicitly in your methodology since it differs across the backbone ablation.

4. Held-out class isolation check

A verification step (not just a manual glance) confirming the held-out class has zero leakage into training/validation for each of the 6 folds trivial to get wrong with a bad path filter, and it silently invalidates your entire OOD claim if it happens.